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Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ-ray spectrometry techniques

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1. INTRODUCTION

The dynamics of neutrons in a nuclear reactor are broadly described by transport theory, which considers spatial, angular, energy, and temporal dependencies. Although this formalism allows predicting the reactor's response to changes in core composition or geometry, the simplifications adopted in the calculations, combined with uncertainties in nuclear data and limitations in the models, result in approximate solutions. Therefore, experimental measurements are essential to validate the models and adjust them to the observed physical reality.

According to theory, the density and spatial distribution of neutron flux inside a reactor are not uniform. This non-uniformity is a consequence of moderation, multiplication, and absorption interactions, which directly depend on the properties of the core materials: fuel, moderator, and reflector. After successive collisions, neutrons tend to concentrate in the central region of the core and disperse into the peripheral regions.

The distribution of neutrons in the core is related to the location of the neutron source, that is, to the arrangement of the fuel elements. In the case of fission, this distribution is influenced by the presence of neutrons themselves and by the interactions with atomic nuclei as they move away from their origin.

From the perspective of diffusion theory, which approximates the transport equation through Fick's Law, the net flux occurs when there are spatial gradients. Thus, neutrons tend to migrate from regions of higher flux density to regions of lower density, analogously to the diffusion of solutes in continuous media.

Understanding this spatial distribution is essential to optimize fuel element design, core geometry, operational safety, and experimental conditions. This work is experimental in nature and aimed to observe, through indirect measurement, the curvature of the neutron flux along the axial axis of the Argonauta Research Reactor at the Nuclear Engineering Institute (IEN/CNEN), under routine operating conditions, with a power of 340 W and a neutron flux on the order of 10^9 n/cm²·s. The methodology employed in this work combined Neutron Activation (Kruger, 1971) and Gamma-Ray Spectrometry techniques.

2. MATERIALS AND METHODS

Dysprosium wires were used; this material was chosen due to its high thermal neutron capture cross section (^{164}Dy, natural abundance 28.26%, capture cross section ~2650 barns). When irradiated, it forms ^{165}Dy , a gamma emitter with a half-life of 8391.6 s and a characteristic peak at 94.7 keV, suitable for gamma spectrometry.

Thirteen Dy wires were distributed on a Lucite (PMMA) ruler, positioned axially between plates 2 and 3 of fuel element No. 6.

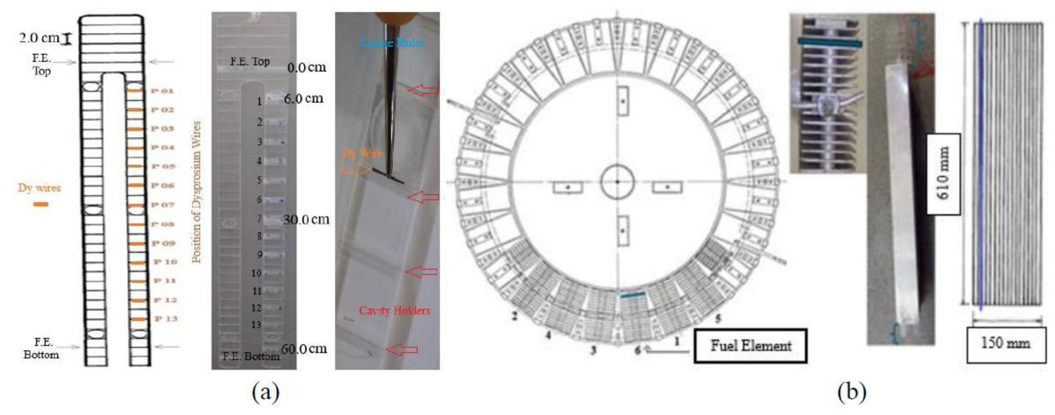

Figure 1: Configuration of dysprosium wires. (a) Positioning of the Dy wires on the Lucite ruler with emphasis on the cavity holders. (b) Insertion position of the Lucite ruler within the reactor core highlighted in blue.

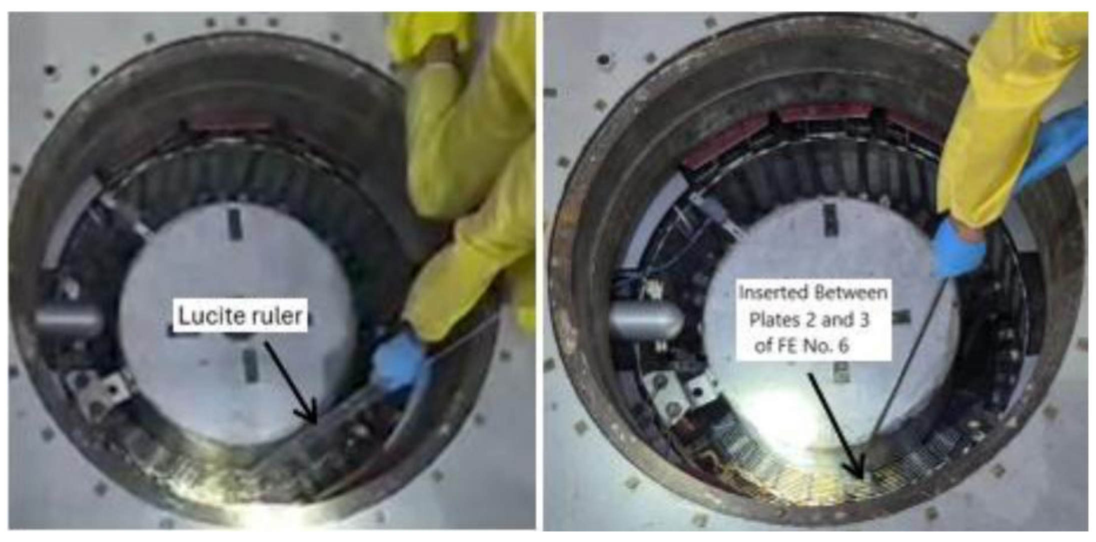

Figure 2: Insertion of the Lucite ruler between fuel plates of FE No.6.

Each wire was 1.1 cm long and 0.711 cm in diameter, made of a Dy–Al alloy (10%). The masses were measured with a high-precision Shimadzu ATY224 balance.

The counts were corrected for dead time and normalized by the mass of each wire, obtaining the specific activity. The HPGe detector (25% efficiency, 1.8 keV resolution) was coupled to the DSPEC jr 2.0 system, with analysis performed using Maestro v7.0 software.

3. RESULTS

The wires were irradiated for 30 min at 340 W, with control rods withdrawn by more than 80%. Each wire was counted at 6 cm from the HPGe detector, which allowed them to be treated as point sources (Bevelacqua, 2004).

The spectra revealed peaks at 94.7 keV, characteristic of ^{165}Dy . The axial flux profile showed a parabolic distribution: maximum concentration in the central region, symmetrically decreasing toward the ends.

This confirms the predictions of diffusion theory in graphite-moderated thermal reactors (Duderstadt & Hamilton, 1976; Lamarsh & Baratta, 1972).

The normalized values showed agreement with the hypothesis of higher flux in the middle region of fuel element No. 6.

Table 1: Normalized Relative Neutron Flux (NRNF) along FE no.6. with Dead-Time Correction (DTC) and Dead-Time (DT).

Dy Wire (No.)	F.E. Position (cm)	NRNF	NRNF (DTC)	DT (%)
1	6.0	0.56±0.01	0.54±0.01	7.37
2	10.0	0.69±0.01	0.67±0.01	8.87
3	14.0	0.78±0.01	0.76±0.01	9.58
4	18.0	0.86±0.01	0.85±0.01	10.34
5	22.0	0.99±0.01	1.00±0.01	11.64
6	26.0	0.98±0.01	0.98±0.01	11.25
7	30.0	0.99±0.01	0.99±0.01	11.08
8	34.0	1.00±0.01	1.00±0.01	11.10
9	38.0	0.95±0.01	0.95±0.01	10.45
10	42.0	0.93±0.01	0.91±0.01	9.88
11	46.0	0.77±0.01	0.74±0.01	8.05
12	50.0	0.75±0.01	0.72±0.01	7.64
13	54.0	0.60±0.01	0.57±0.01	6.07

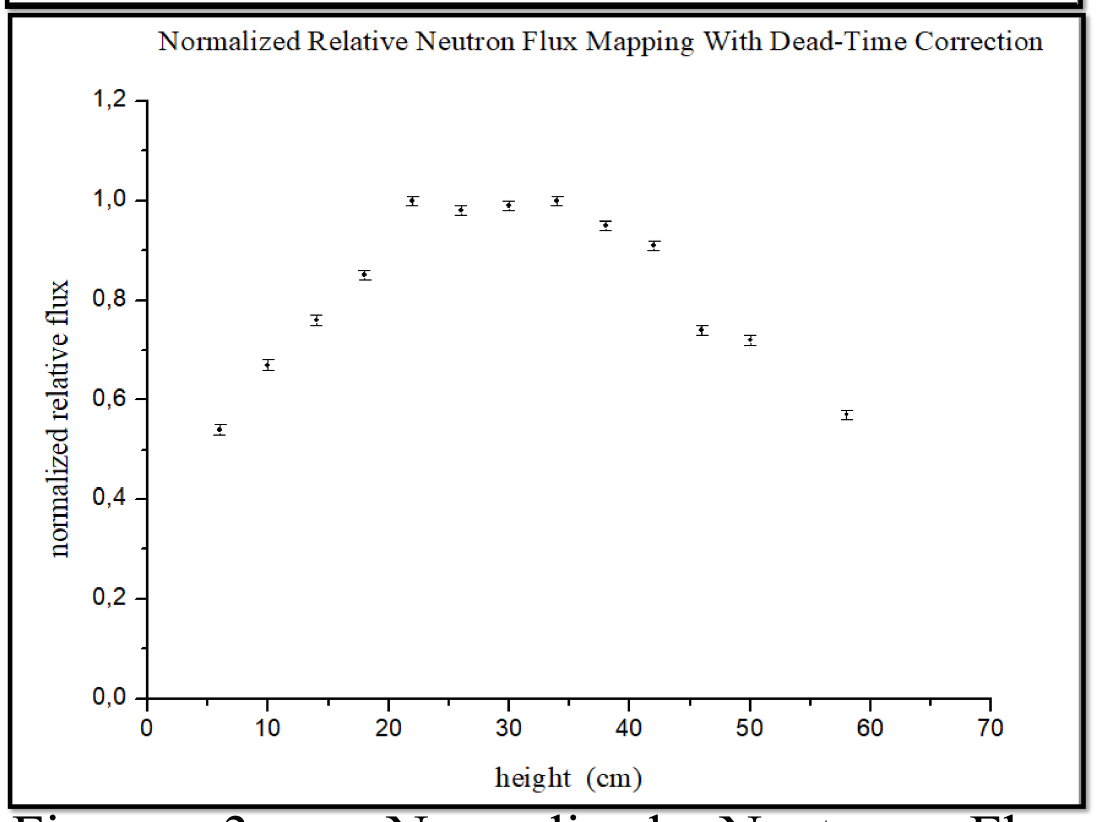

Figure 3 - Normalized Neutron Flux Mapping with Dead-Time Correction and Error Bars.

4. CONCLUSIONS

The experimental methodology made it possible to qualitatively map the axial neutron flux between plates 2 and 3 of fuel element No. 6 in the Argonauta reactor, operating routinely at 340 W.

The activation of Dy wires combined with gamma spectrometry using a high-resolution HPGe detector enabled indirect observation of flux behavior, confirming theoretical predictions of central concentration and peripheral dispersion.

The obtained profile demonstrates the feasibility of the technique as a method for qualitative assessment of flux distribution. For future work, it is recommended to expand the mapping to other regions of the core, including the radial plane, and to compare the results with computational simulations.